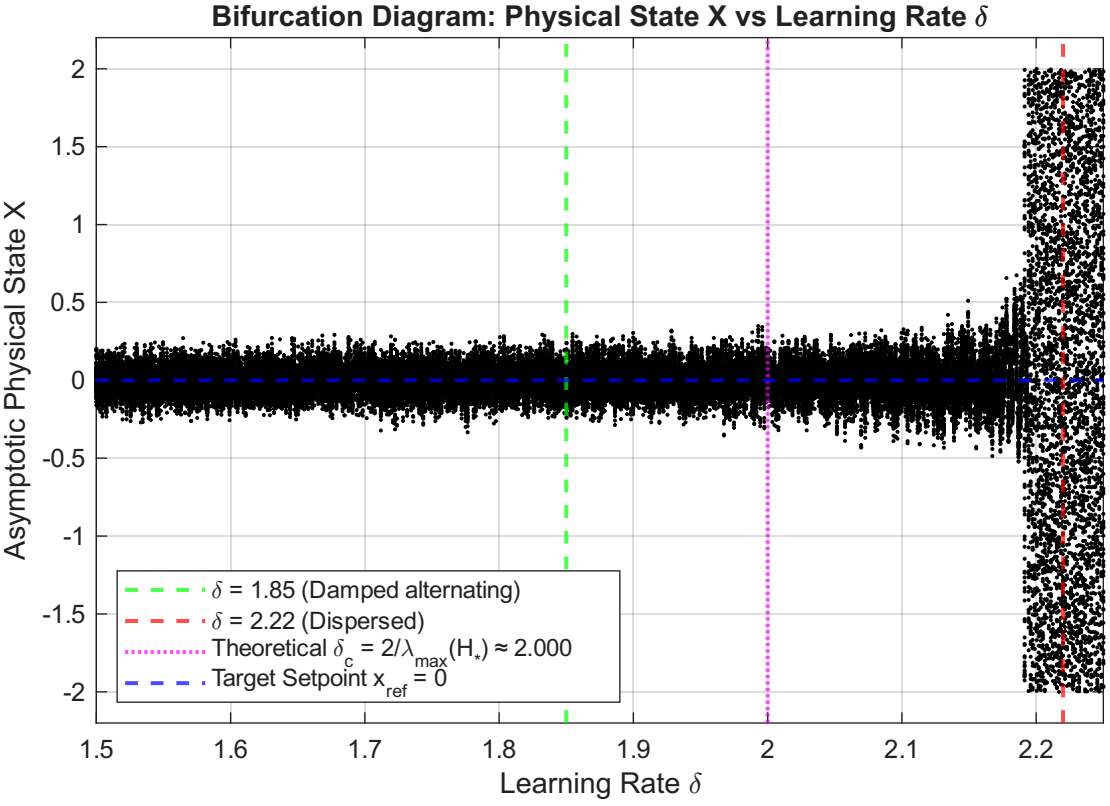

Bifurcation Diagram: Physical State X vs Learning Rate δ
Asymptotic Physical State X
Learning Rate δ
2
1.5
1
0.5
0
-0.5
-1
-1.5
-2
1.5
1.6
1.7
1.8
1.9
2
2.1
2.2
δ = 1.85 (Damped alternating)
δ = 2.22 (Dispersed)
Theoretical δc = 2/λmax(H*) ≈ 2.000
Target Setpoint xref = 0